

E3

Site Analysis

Building typology and massing play an important role in responding to seasonal solar conditions in New York. The form of the building is shaped to create deep overhangs and angled surfaces that respond to the sun's path during the hottest periods of the year. As shown in the solar study, the massing is positioned to block high summer sun, allowing the building envelope and projecting volumes to act as passive shading devices. These overhangs reduce direct solar gain on exposed façades and outdoor surfaces, helping to lower interior heat loads and reduce cooling demand.

The orientation and curvature of the massing are informed by solar analysis, allowing the building to strategically intercept sunlight during peak summer hours while still permitting daylight penetration during other times of the year. By integrating solar performance directly into the building's typological form, the architecture itself becomes part of the environmental control system rather than relying solely on mechanical solutions.

This approach demonstrates how massing, overhang, and building geometry can work together to mitigate heat gain during New York's hottest months while improving overall energy performance and occupant comfort.

Images show summer solstice massing studies:

